OSPF LAB

Router 1

R1

```
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.0.0.0 0.255.255.255 a 1
R1(config-router)#network 20.0.0.0 0.255.255.255 a 1
R1(config-router)#exit
R1(config)#
*Oct  6 14:02:47.044: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/1 from
 LOADING to FULL, Loading Done
R1(config)#
*Oct  6 14:05:33.903: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R1(config)#
R1(config)#do sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:33    20.0.0.2        Ethernet0/1
R1(config)#do sh ip ospf database

            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         604         0x80000005 0x0017B8 2
2.2.2.2         2.2.2.2         460         0x80000006 0x003959 2
3.3.3.3         3.3.3.3         420         0x80000005 0x008606 2

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
20.0.0.1        1.1.1.1         604         0x80000001 0x00F81C
30.0.0.1        2.2.2.2         460         0x80000001 0x00AC52
R1(config)#do sh ip ospf database

            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1356        0x80000005 0x0017B8 2
2.2.2.2         2.2.2.2         1212        0x80000006 0x003959 2
3.3.3.3         3.3.3.3         1172        0x80000005 0x008606 2

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
20.0.0.1        1.1.1.1         1356        0x80000001 0x00F81C
30.0.0.1        2.2.2.2         1212        0x80000001 0x00AC52
R1(config)#
```

Router 2

R2

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 20.0.0.0 0.255.255.255 a 1
R2(config-router)#network 30.0.0.0 0.255.255.255 a 1
*Oct  6 14:02:47.044: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R2(config-router)#network 30.0.0.0 0.255.255.255 a 1
R2(config-router)#exit
R2(config)#
*Oct  6 14:05:12.599: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/1 from
 LOADING to FULL, Loading Done
R2(config)#
*Oct  6 14:05:48.630: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R2(config)#do sh ip neighbor
                  ^
% Invalid input detected at '^' marker.

R2(config)#do sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:30    30.0.0.2        Ethernet0/1
1.1.1.1           1   FULL/DR         00:00:33    20.0.0.1        Ethernet0/0
R2(config)#do sh ip ospf database

            OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1214        0x80000005 0x0017B8 2
2.2.2.2         2.2.2.2         1067        0x80000006 0x003959 2
3.3.3.3         3.3.3.3         1027        0x80000005 0x008606 2

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
20.0.0.1        1.1.1.1         1214        0x80000001 0x00F81C
30.0.0.1        2.2.2.2         1067        0x80000001 0x00AC52
R2(config)#
```

Router 3

R3

```
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 30.0.0.0 0.255.255.255 a 1
R3(config-router)#network 40.0.0.0 0.255.255.255 a 1
R3(config-router)#
*Oct  6 14:05:12.599: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/1 from
 LOADING to FULL, Loading Done
R3(config-router)#exit
R3(config)#do sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:35    30.0.0.1        Ethernet0/1
R3(config)#do sh
*Oct  6 14:05:39.537: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R3(config)#do sh ip ospf database

            OSPF Router with ID (3.3.3.3) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         181         0x80000005 0x0017B8 2
2.2.2.2         2.2.2.2         35          0x80000006 0x003959 2
3.3.3.3         3.3.3.3         24          0x80000004 0x008805 2

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
20.0.0.1        1.1.1.1         181         0x80000001 0x00F81C
30.0.0.1        2.2.2.2         35          0x80000001 0x00AC52
R3(config)#do sh ip ospf database

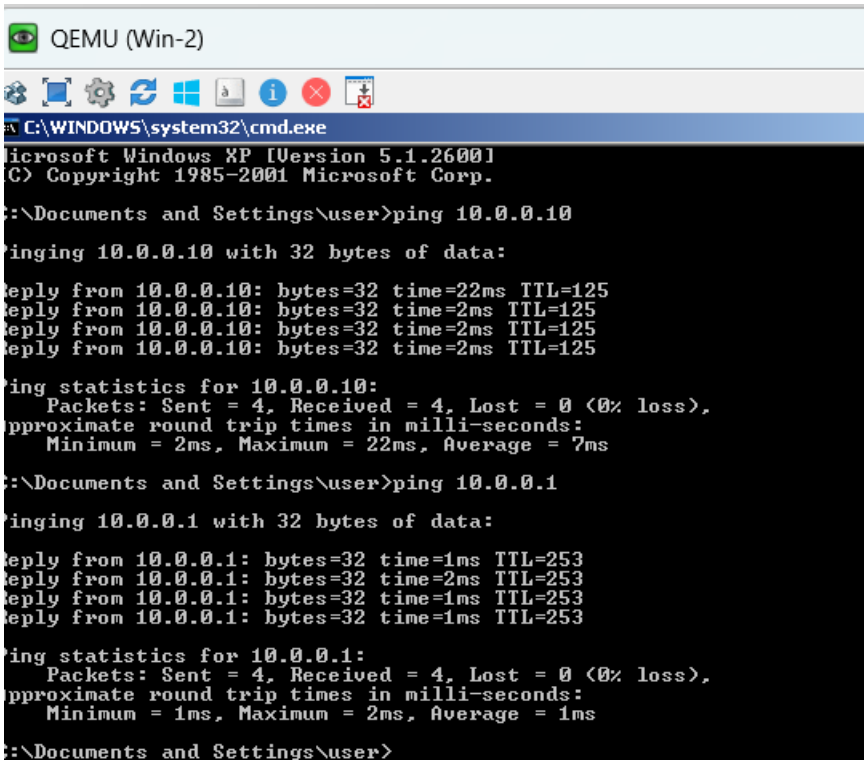
            OSPF Router with ID (3.3.3.3) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         843         0x80000005 0x0017B8 2
2.2.2.2         2.2.2.2         696         0x80000006 0x003959 2
3.3.3.3         3.3.3.3         654         0x80000005 0x008606 2

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
20.0.0.1        1.1.1.1         843         0x80000001 0x00F81C
30.0.0.1        2.2.2.2         696         0x80000001 0x00AC52
R3(config)#
```


QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
icrosoft Windows XP [Version 5.1.2600]
C) Copyright 1985-2001 Microsoft Corp.
:\Documents and Settings\user>ping 10.0.0.10
inging 10.0.0.10 with 32 bytes of data:
eply from 10.0.0.10: bytes=32 time=22ms TTL=125
eply from 10.0.0.10: bytes=32 time=2ms TTL=125
eply from 10.0.0.10: bytes=32 time=2ms TTL=125
eply from 10.0.0.10: bytes=32 time=2ms TTL=125
ing statistics for 10.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 22ms, Average = 7ms
:\Documents and Settings\user>ping 10.0.0.1
inging 10.0.0.1 with 32 bytes of data:
eply from 10.0.0.1: bytes=32 time=1ms TTL=253
eply from 10.0.0.1: bytes=32 time=2ms TTL=253
eply from 10.0.0.1: bytes=32 time=1ms TTL=253
eply from 10.0.0.1: bytes=32 time=1ms TTL=253
ing statistics for 10.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
:\Documents and Settings\user>